



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



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Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

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Basic Equations

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Finite Volume Method

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Practical Implementation

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Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

- in collisional fluids, large enough to communicate local conditions, requires interactions

$$\ell \gg \lambda$$

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial\mathbf{t}$ for any measurable quantity Q
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via $D/D\mathbf{t}$ as the rate of change
- *Translating Relation*: add convective derivative term to account for movement of fluid element

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

■ *Derived Values:*

- viscosity η (scalar)
- vorticity $\boldsymbol{\omega}$ (vector)

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Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)
 - ▶ $10 \lesssim R \lesssim 40$: Separation Zone (stable and stationary vortices)
- $R > 1000$: Turbulent Flow (stochastic and chaotic stream)
 - ▶ $40 \lesssim R \lesssim 4000$: Transitional Kármán Street (vortex shedding)
 - ▶ $R > 4000$: Kolmogorov Energy Cascade (fully developed phase)

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

- $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)
- $M \approx 1$: Transonic (local excess of sound speed, weak shocks create drag)
- $M > 1$: Supersonic (thin front of discontinuity, bow shock heating)
- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- *Linear Advection*: velocity field $\mathbf{v}$ is free of divergence for incompressible fluids

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} = 0$$

- *Nonlinear Advection*: velocity field $\mathbf{u}$ is carried by itself instead of fixed background flow

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = 0$$

Linear Advection

Nonlinear Advection

Stochastic Transport

- *Diffusion*: kinematic viscosity ν describes spread due to random motion

$$\frac{\partial u}{\partial t} = \nu \nabla^2 u$$

- *Bateman-Burgers Equation*: combines shock generation with counteracting diffusive term

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = \nu \nabla^2 u$$

Diffusion

Bateman-Burgers Equation

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

- Gravity: $\nabla^2 \Phi = 4\pi G\rho$
- Electrostatics: $\nabla^2 V = -\rho/\epsilon$
- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla)u)$

Laplace Equation

Poisson Equation

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Lid Driven Cavity Flow

Pressure Driven Channel Flow

Nonuniform Periodic Flow

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Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems
- Drawbacks:
 - struggles with complex geometries
 - strong numerical diffusion at low orders
 - inflexible rectilinear grid with fixed resolution
 - mass and energy can leak due to numerical errors

Integration Schemes

■ a

Discretization

■ a

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Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain in integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:
 - inherently strictly conservative for total mass and energy
 - unstructured grid handles complex geometries well
- Drawbacks:
 - mathematically complex at higher order accuracy

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■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ Regimes:

- | | |
|--|--|
| ■ <i>Plasma</i> : Magnetohydrodynamics (MHD) | ■ <i>Degeneracy</i> : Quantum Effects |
| ■ <i>Relativity</i> : Relativistic Hydrodynamics (RHD) | ■ <i>Collisionless</i> : Initial Distributions |

■ Applications:

- | | | |
|---------------------|-------------------|--------------------|
| ■ Stellar Evolution | ■ Accretion Disks | ■ Cosmic Filaments |
|---------------------|-------------------|--------------------|

Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)